

# Basic Two Pointers (Same Direction)

## 1. Two Sum II – Input Array Is Sorted

- Use left + right pointer
- Move based on sum comparison

## 2. Remove Duplicates from Sorted Array

- Slow & fast pointer

## 3. Remove Element

- Overwrite unwanted elements

## 4. Move Zeroes

- Shift non-zero forward

## 5. Valid Palindrome

- Compare left/right characters

## 6. Merge Sorted Array

- Fill from end using two pointers

# Opposite Direction Pointers

## 7. Container With Most Water

- Classic left/right shrink problem

## **8. 3Sum**

- Sort + fix one element + two pointers

## **9. 3Sum Closest**

## **10. 4Sum**

## **11. Sort Colors (Dutch National Flag)**

# **Sliding Window (Dynamic Two Pointers)**

12. Longest Substring Without Repeating Characters

13. Minimum Size Subarray Sum

14. Permutation in String

15. Find All Anagrams in a String

16. Longest Repeating Character Replacement

# **Fast & Slow Pointer (Cycle Detection)**

17. Linked List Cycle

18. Linked List Cycle II

19. Middle of the Linked List

20. Happy Number

# **Advanced Two Pointer**

21. Trapping Rain Water

22. Backspace String Compare

23. Boats to Save People

24. Squares of a Sorted Array

## If You're Preparing in Order

### Beginner Level

- Remove Duplicates
- Move Zeroes
- Valid Palindrome
- Two Sum II

### Medium Level

- Container With Most Water
- 3Sum
- Longest Substring Without Repeating Characters
- Sort Colors

### Advanced

- Trapping Rain Water
- 4Sum
- Linked List Cycle II

# Most Important (Must Master)

1. Two Sum II
2. Remove Duplicates
3. Valid Palindrome
4. Container With Most Water
5. 3Sum
6. Longest Substring Without Repeating Characters

If you master these → you understand 80% of two pointer logic.